

1. Rubber's Entropy

Rubber consists of long-chain molecules. With no force applied, the rubber molecules are quite coiled up, and the average end-to-end distance is short. With a force applied (along horizontal axis), the molecules become more aligned with the direction of the force, and end-to-end distance is large reducing entropy.

A rubber molecule can be modeled in one dimension as a chain consisting of a series of $N = N_+ + N_-$ links, where N_+ links point in the $+x$ direction and N_- links point in the $-x$ direction, respectively.

If the length of one link in the chain is a ,

(a) Show that the length L of the chain is

$$L = a(N_+ - N_-)$$

$$\begin{aligned} L &= aN \\ &= a(N_+ - N_-). \end{aligned}$$

(b) The number of ways $\Omega(L)$ of arranging the links to achieve a length L can be written as

$$\Omega(L) = \frac{N!}{N_+!N_-!}.$$

Show that the entropy, S can be written approximately as

$$S = Nk_B \left[\ln 2 - \frac{L^2}{2N^2a^2} \right]$$

when $L \ll Na$.

$$\begin{aligned} \frac{S}{k_B} &= \ln \Omega \\ &= \ln \left(\frac{N!}{N_+!N_-!} \right) \\ &= \ln N! - \ln N_+! - \ln N_-! \\ &= N \ln N - N - (N_+ \ln N_+ - N_+) - (N_- \ln N_- - N_-) \\ &= N \ln N - N - \left(\frac{1}{2} \left(N + \frac{L}{a} \right) \ln \frac{1}{2} \left(N + \frac{L}{a} \right) - \left(\frac{1}{2} \left(N + \frac{L}{a} \right) \right) \right) - \\ &\quad \left(\frac{1}{2} \left(N - \frac{L}{a} \right) \ln \frac{1}{2} \left(N - \frac{L}{a} \right) - \frac{1}{2} \left(N - \frac{L}{a} \right) \right) \\ &= \ln N! - \ln \left[\left(\frac{1}{2} N + \frac{L}{a} \right)! \right] - \ln \left[\left(\frac{1}{2} N - \frac{L}{a} \right)! \right] \\ &= N \ln N - \frac{N + L/a}{2} \ln \left(\frac{N + L/a}{2} \right) - \frac{N - L/a}{2} \ln \left(\frac{N - L/a}{2} \right) \end{aligned}$$

$$\begin{aligned}
&= N \ln N - \frac{N + L/a}{2} \left(\ln \frac{N}{2} + \ln \left(1 + \frac{L}{Na} \right) \right) - \frac{N - L/a}{2} \left(\ln \frac{N}{2} + \ln \left(1 - \frac{L}{Na} \right) \right) \\
&= N \ln N - N \ln \left(\frac{N}{2} \right) - \frac{N}{2} \left[\left(1 + \frac{L}{Na} \right) \ln \left(1 + \frac{L}{Na} \right) + \left(1 - \frac{L}{Na} \right) \ln \left(1 - \frac{L}{Na} \right) \right] \\
&\approx N \ln 2 - \frac{N}{2} \left[\left(\frac{L}{Na} + \frac{(L/Na)^2}{2} \right) + \left(-\frac{L}{Na} + \frac{(L/Na)^2}{2} \right) \right], \\
&\quad \text{using rough Taylor series approximation} \\
&= N \ln 2 - \frac{N}{2} \left(\frac{L^2}{N^2 a^2} \right) \\
&= N \ln 2 - \frac{L^2}{2Na^2} \\
S &= Nk_B \left[\ln 2 - \frac{L^2}{2N^2 a^2} \right].
\end{aligned}$$

2. Black Hole Thermodynamics (A simple model)

We are given expressions for the entropy, temperature, and event horizon radius of a black hole:

$$\begin{aligned}
S &= \frac{k_B c^3}{4\hbar G} A, \\
T &= \frac{\hbar c^3}{8\pi k_B G M}, \\
R &= \frac{2GM}{c^2}.
\end{aligned}$$

(a) Show that S can be written in the form

$$S = \frac{4\pi k_B G}{\hbar c} M^2.$$

$$\begin{aligned}
S &= \frac{k_B c^3}{4\hbar G} A, \text{ where } A = 4\pi R^2 = \frac{16\pi G^2 M^2}{c^4} \\
&= \frac{k_B c^3}{4\hbar G} \cdot \frac{16\pi G^2 M^2}{c^4} \\
&= \frac{4\pi k_B G}{\hbar c} M^2.
\end{aligned}$$

(b) If mass losses, show that

$$dS = \frac{8\pi k_B G}{\hbar c} m dm.$$

$$\begin{aligned}
\frac{dS}{dm} &= \frac{8\pi k_B G}{\hbar c} m \\
dS &= \frac{8\pi k_B G}{\hbar c} m dm.
\end{aligned}$$

- (c) If the initial mass of the black hole is M and final mass of it is M' , show that the heat energy lost is given by

$$Q = \Delta M c^2.$$

$$\begin{aligned} dS &= \frac{Q}{T} \\ Q &= T dS \\ &= \frac{8\pi k_B G}{\hbar c} M \Delta M \cdot \frac{\hbar c^3}{8\pi k_B G M} \\ &= (\Delta M) c^2 \end{aligned}$$

- (d) Show that the heat capacity of the black hole C is given by

$$C = -\frac{8\pi k_B G}{\hbar c} M.$$

$$\begin{aligned} C &= T \frac{dS}{dT} \\ &= \frac{\hbar c^3}{8\pi k_B G M} \cdot \frac{8\pi k_B G M dM}{\hbar c} \cdot -\frac{8\pi k_B G M^2}{\hbar c^3 dM} \\ &= -\frac{8\pi k_B G}{\hbar c} M^2. \end{aligned}$$

My derivation yields the heat capacity of the black hole to be $C = -\frac{8\pi k_B G}{\hbar c} M^2$, not $C = -\frac{8\pi k_B G}{\hbar c}$.

What is the significance of negative heat capacity?

Negative heat capacity implies that when energy is added to a black hole, its temperature is lowered, while when energy is removed from a black hole, its temperature increases.